

to control the flow of current within the chips. An unwanted flow of current is called current leakage. This makes the transistors move from a pure On or Off state, to a fuzzy state. Another problem is manufacturing costs. Companies have to keep decreasing the cost per transistor so that they can offer better processors at the same, or marginally higher cost.

Scientists are looking at various methods to extend Moore's law—some are evolutionary, few revolutionary. All these can be classified into the following groups: looking for alternate materials other than silicon; using light instead of electricity to transfer data within the chip; shrinking the existing technology; completely different fabrication methods that involve polymers; looking at other forms of computing such as quantum computing, DNA computing. Let's look at some notable efforts.

High-k metal

Transistors act like On/Off switches. The gate (gate electrode), is a part of the transistor that dictates whether the transistor is in the On or Off state. Below the gate, is the gate dielectric that acts like an insulator, preventing the flow of electricity. For the past 30 years, silicon dioxide has been used as the gate dielectric. However, reducing the thickness of the silicon dioxide gate dielectric results in higher current leakage. Hence, researchers are on the lookout for alternate materials with better insulating properties. Intel has developed a new class of materials, called High-k to replace silicon dioxide as the gate dielec-

tric. High-k stands for 'high dielectric constant' and is a measure of how much charge a material can hold. Air is used as the reference point for this constant, i.e. air has a 'k' of 1.0, and similarly silicon dioxide has a 'k' of 3.9. Intel has discovered materials such as hafnium dioxide (HfO_2), zirconium dioxide (ZrO_2) and titanium dioxide (TiO_2) that have higher dielectric constants than silicon dioxide. However, the newly discovered high-k materials won't work with the existing gate electrodes that're made of polysili-



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Intel will use High-k along with other innovations, such as strained silicon and tri-gate transistors, to extend transistor scaling and Moore's Law

con. So, Intel developed the right type of gate electrode for the new high-k gate dielectric. The new gate electrode will be made of metal.

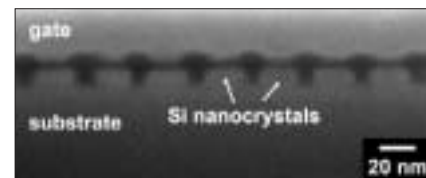
When a material with high 'k' value is used as a gate dielectric, it results in transistors with higher capacitance. It's very easy to distinguish between the On and Off state when a transistor has high capacitance. This also results in faster transistors. Also, high-k materials reduce gate leakages a 100 times, which trans-

lates to chips running at much lower temperatures. Thus, the high-k/metal gate technology removes the major block towards the continuing of Moore's law. Intel made this announcement in November 2003, after five years of research. It claims that the process is on track for production in 2007.

Self-assembly

In December 2003, two IBM researchers, Chuck Black and Kathryn Guarini, announced that they have built a Flash memory device using self-assembly techniques. Self-assembly is a subset of nanotechnology that refers to the ability of certain objects to autonomously arrange themselves into patterns.

The most important feature about the IBM development is that it's compatible with the existing manufacturing tools, thus obviating the need for expensive tooling changes. "As components in information technology products continue to shrink toward the molecular scale, self-assembly techniques could be used to



Magnified image of the FLASH memory device created by IBM using self assembly

enhance lithographic methods," said Dr T C Chen, Vice President of science and technology at IBM Research.

Currently, photolithography is used to etch connections on the silicon wafer with a combination of light and a mask—akin to a stencil. However, at atomic scales, it is not possible to form precise structures with this process. IBM researchers used polymer molecules that organise themselves autonomously into precise hexagonal patterns. These patterns have small holes with a diameter of 20 nanometres (nm) and the holes are separated by 40 nm. This is used as a stencil to reproduce a similar pattern in silicon dioxide, after which the polymer is removed and is no longer needed. Now,

But is Moore's Law Relevant, Anymore?

Moore's Law has put the chip manufacturers on an aggressive road map. Applications aren't demanding powerful processors anymore; it seems to be the other way round. Companies are foreseeing applications for the processors that are yet to be released. Over the years, while the transistors in a chip seem to be increasing exponentially, the end-user experience doesn't seem to be reflecting that trend. Without a doubt, there is a need for more powerful processors and companies do have to keep developing them, but we have reached a stage where there is no need for such high acceleration to move forward. "There is very little in terms of mainstream software that is coming out anytime soon that needs more perform-

ance on the desktop," said Mark Margevicius, Research Director at Gartner Inc. While many wouldn't agree with him, they would agree that there is a sizeable population using their PC only for light work such as word processing, browsing, etc. Is anything being made to improve their experience?

Using multiple processors instead of one powerful processor is another direction to go. Google, for example, has one of the largest processor farms. When the 130W consuming Itanium was released, Google refused to buy any of it, citing its power consumption. Many of the systems used by Google are Celeron-based. Maybe it's time to have a second look at Moore's Law, maybe not!